

Practice Lab:

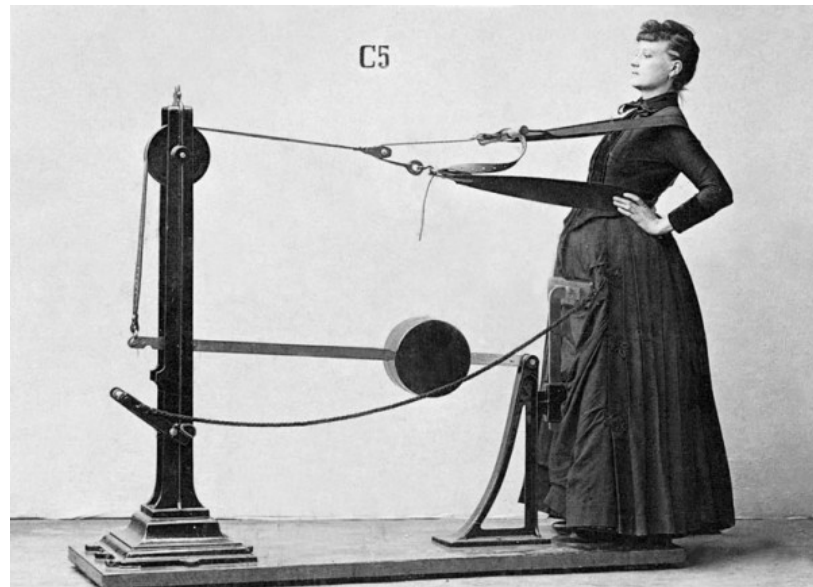
DSE Core, alter, create, and examine keyspaces

- This Practice Lab has no specific preceding Discussion Unit. A 101-level expertise is required on given terms; data center, keyspace, (and others, as listed on the Notes page below).
- This Practice Lab is meant to address two issues:
 - After you create a new DSE cluster, a number of the system keyspaces exist with SimpleStrategy, and should be altered to NetworkTopologyStrategy.
 - If you inherit a cluster with down nodes, and seek to repair same, how to determine if data is at risk.

Challenge 1: Create a 2 DC, 1 Node per DC Cluster

Prerequisites:

- Instructions are provided for Linux
- We will operate below recommended settings; 1 OS node only, minimum 8 GB RAM, 12 GB RAM preferred
- That you have downloaded a DSE Tar ball
- That you are familiar with a DSE Tar ball install; cassandra.yaml, and similar
- *All prior DSE software is shut down, resources reclaimed*
- All work done as 'root'



Challenge 1: Files, directories



- Unpack the DSE Tar ball in /opt/node0
- Copy these contents into
 - /opt/node1 # aka “node1”
 - /opt/node2 # aka “node2”
- Under node1, node2, make a ./data folder
- Under each ./data, make the folders
 - ./cdc_raw
 - ./commitlog
 - ./data # results in ./data/data
 - ./hints
 - ./saved_caches

Challenge 1: cassandra.yaml

In the cassandra.yaml in each of node1/node2, change:

- cluster_name: 'my_cluster'
- endpoint_snitch:
org.apache.cassandra.locator.GossipingPropertyFileSnitch
- - seeds: "127.0.0.1, 127.0.0.2"
(above, properly indented, same on both)
- listen_address: 127.0.0.1 # node 1
- listen_address: 127.0.0.2 # node 2
- native_transport_address: 127.0.0.1 # node 1
- native_transport_address: 127.0.0.2 # node 2



DseSimpleSnitch
PropertyFileSnitch
GossipingPropertyFileSnitch

Challenge 1: cassandra.yaml

In the cassandra.yaml in each of node1/node2, change:

- `commitlog_directory: /opt/dse/node1/data/commitlog` # node 1
- `commitlog_directory: /opt/dse/node2/data/commitlog` # node 2

(Each below different for `node1`, `node2`)

- `cdc_raw_directory: /opt/dse/node1/data/cdc_raw`
- `hints_directory: /opt/dse/node1/data/hints`
- `saved_caches_directory: /opt/dse/node1/data/saved_caches`
- `data_file_directories:`
 - `/opt/dse/node1/data/data`

Challenge 1: cassandra-env.sh

In the cassandra-env.sh in each of node1/node2, change:

- The first two will be commented, uncomment and set to,

```
MAX_HEAP_SIZE="2048M"    # Same both nodes  
HEAP_NEWSIZE="200M"     # Same both nodes
```

- JMX_PORT="7199" # Node 1
- JMX_PORT="7299" # Node 2



Challenge 1: logback.xml

In the logback.xml in each of node1/node2, **[globally]** change:

```
${cassandra.logdir}  
to  
/opt/dse/node1/    # node 1  
/opt/dse/node2/    # node 2
```

Generally, this change is made 8 times.



Challenge 1: `cassandra-rackdc.properties`



The node 1 (rackdc) properties file is fine.
You only need to edit this file on node 2.

- Make this change-
`dc=dc2`

Challenge 1: Boot both DSE nodes

- 3 Linux terminal windows-
- In window 1, `cd /opt/dse/node1/bin`
- In window 2, `cd /opt/dse/node2/bin`
- In both windows above,
`./dse cassandra -f -R`
- In window 3,
`cd /opt/dse/node1/bin`
`nodetool ring` `# repeat this command`

Challenge 1: Success

A successful exercise outputs-

```
# ./nodetool ring
```

```
Datacenter: dc1
```

```
=====
```

Address	Rack	Status	State	Load	Owns	Token
127.0.0.1	rack1	Up	Normal	87.99 KiB	?	6676891273543202389

```
Datacenter: dc2
```

```
=====
```

Address	Rack	Status	State	Load	Owns	Token
127.0.0.2	rack1	Up	Normal	40.39 KiB	?	-3406290366391324992

Challenge 2: Changing system keyspaces

```
# cqlsh
```

```
Connected to my_cluster at 127.0.0.1:9042.
```

```
[cqlsh 5.0.1 | Cassandra 4.0.0.2284 | DSE 6.0.0 |  
  CQL spec 3.4.5 | DSE protocol v2]
```

```
Use HELP for help.
```

```
cqlsh> use system_schema;
```

```
cqlsh:system_schema> describe tables;
```

tables	triggers	views	keyspaces	dropped_columns
functions	aggregates	indexes	types	columns

```
cqlsh:system_schema> select * from keyspaces;
```

Challenge 2: Which keyspaces need change ?

```
cqlsh:system_schema> select * from keyspaces;
```

keyspace_name	...	replication
system_auth	...	{'class': 'org.apache.cassandra.locator.SimpleStrategy', 'replication_factor': '1'}
system_schema	...	{'class': 'org.apache.cassandra.locator.LocalStrategy'}
dse_system_local	...	{'class': 'org.apache.cassandra.locator.LocalStrategy'}
dse_system	...	{'class': 'org.apache.cassandra.locator.EverywhereStrategy'}
dse_leases	...	{'class': 'org.apache.cassandra.locator.SimpleStrategy', 'replication_factor': '1'}
solr_admin	...	{'class': 'org.apache.cassandra.locator.EverywhereStrategy'}
system_distributed	...	{'class': 'org.apache.cassandra.locator.SimpleStrategy', 'replication_factor': '3'}
system	...	{'class': 'org.apache.cassandra.locator.LocalStrategy'}
dse_perf	...	{'class': 'org.apache.cassandra.locator.SimpleStrategy', 'replication_factor': '1'}
system_traces	...	{'class': 'org.apache.cassandra.locator.SimpleStrategy', 'replication_factor': '2'}
dse_security	...	{'class': 'org.apache.cassandra.locator.SimpleStrategy', 'replication_factor': '1'}

(11 rows)

```
cqlsh:system_schema>
```

Challenge 2: ALTER KEYSPACE

```
# cqlsh
alter keyspace system_auth
  with replication =
    {
      'class' : 'NetworkTopologyStrategy',
      'dc1'   : 1,
      'dc2'   : 1
    }
  and
  durable_writes = 'true' ;
```

Warnings :

When increasing replication factor you need to run a non-incremental repair on all nodes to distribute the data (nodetool repair -pr).

Challenge 2: nodetool repair

```
# nodetool -h 127.0.0.1 repair -pr
[2018-06-23 ... ] Starting repair command #1 ... repairing keyspace dse_system with ...
[2018-06-23 ... ] Repair session ... finished
[2018-06-23 ... ] Repair completed successfully
[2018-06-23 ... ] Repair command #1 finished in 0 seconds

[2018-06-23 ... ] Replication factor is 1. No repair is needed for keyspace 'dse_leases'

[2018-06-23 ... ] Starting repair command #2 ... repairing keyspace solr_admin with ...
[2018-06-23 ... ] Repair session ... finished
[2018-06-23 ... ] Repair completed successfully
[2018-06-23 ... ] Repair command #2 finished in 0 seconds

[2018-06-23 ... ] Replication factor is 1. No repair is needed for keyspace 'dse_perf'

[2018-06-23 ... ] Starting repair command #3 ... repairing keyspace system_traces with ...
[2018-06-23 ... ] Repair session ... finished
[2018-06-23 ... ] Repair completed successfully
[2018-06-23 ... ] Repair command #3 finished in 0 seconds

[2018-06-23 ... ] Replication factor is 1. No repair is needed for keyspace 'dse_security'

[2018-06-23 ... ] Starting repair command #4 ... repairing keyspace system_auth with ...
[2018-06-23 ... ] Repair session ... finished
[2018-06-23 ... ] Repair completed successfully
[2018-06-23 ... ] Repair command #4 finished in 0 seconds
```

Challenge 2: Confirm change

```
cqlsh:system_schema> select * from keyspaces;
```

```
keyspace_name | ... | replication
```

```
-----+-----+-----  
system_auth | ... | {'class': 'org ... NetworkTopologyStrategy', 'dc1': '1', 'dc2': '1'}  
...
```

Challenge 3: Will [you] cause data loss ? [Discuss]

node down , N		nodetool decommission nodetool netstats	
node down, Y 	Simple Strategy	RF > # of down nodes	
		RF <= # of down nodes	
	Network Topology Strategy	RF spans DCs, At least 1 DC not affected	
		RF > # of down nodes AT LEAST 1 DC	
		RF spans all DCs and RF <= # of down nodes IN EACH DC <i>Most likely ? Restore from backup</i>	

Practice Lab:

Lessons Learned



